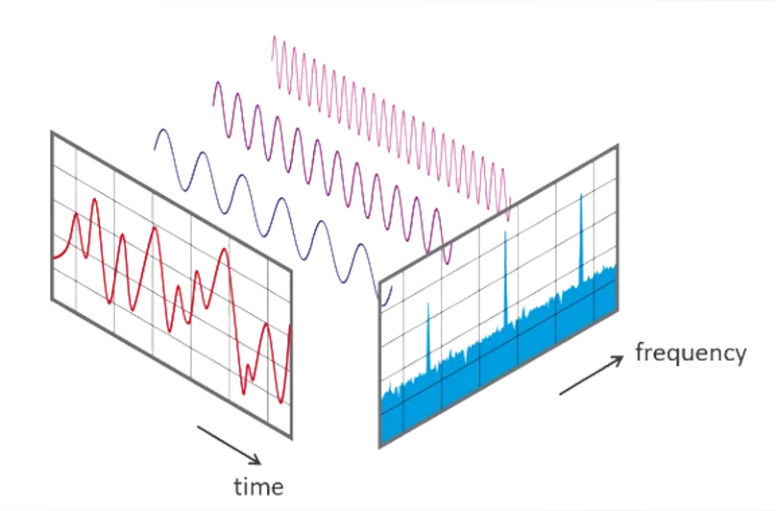
FREQUENCY-DOMAIN METHODS

Statistical Validation of Computer Models via Frequency-Domain Methods



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01 · MOTIVATION

Engineering decisions require trustworthy computer models.

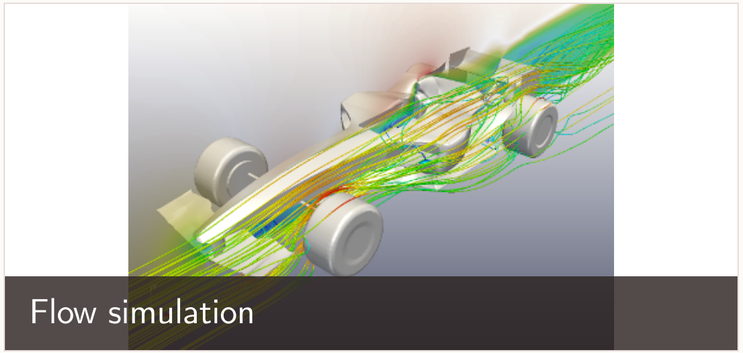
Computer simulations support scientific discovery and engineering decisions.

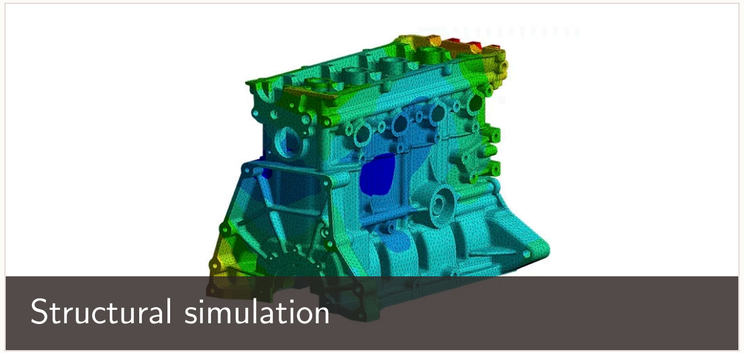
Before practical use, we must validate their reliability.

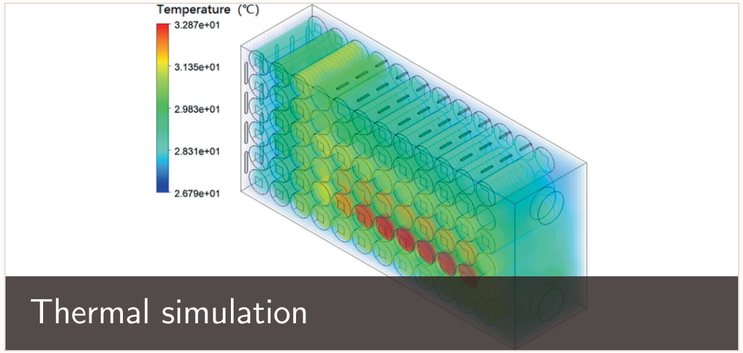
Validation asks whether the computer input-output relation agrees with the physical process.

Because physical observations are finite and noisy, visual comparison alone is not enough.

We therefore need a calibrated p-value and a global or subdomain decision.


Flow simulation


Structural simulation



Thermal simulation

Validation connects simulation outputs to physical evidence.

02 · INTUITION

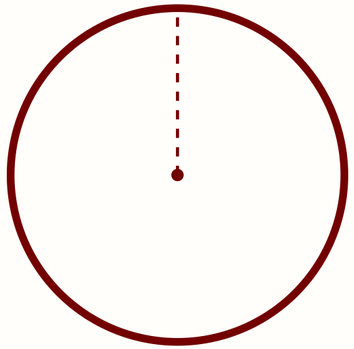
A perfect circle as a validation problem.

Can you draw a perfect circle?

Scan to play

Your drawing is the experiment.

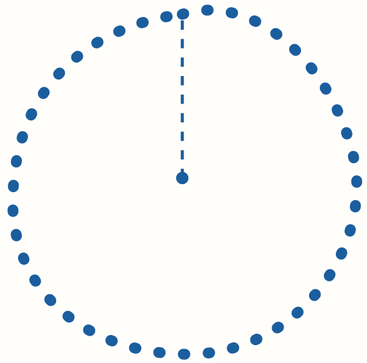
Computer model



ideal circle

vs.

Physical observation



one drawing

Is the observed drawing statistically consistent with the perfect circle?

From an ideal circle to a physical observation.

03 · FORMAL SETUP

The validation problem, formalized as a hypothesis test.

1. **Computer model.** $y^s : \Omega \to \mathbb{R},\ \Omega \subset \mathbb{R}^d$ — a known, deterministic function: the simulator itself, or a fast surrogate trained on it. We call y^s the computer model, or simply the simulator.
2. **Physical process model.** $y_i = f(x_i) + e_i,\ i = 1, \ldots, n$ — the physical data follow this non-parametric model. f, the physical process, is unknown; it is what y^s is validated against.
3. **Measurement noise.** $e_i \overset{\text{i.i.d.}}{\sim} (0, \sigma^2)$ — independent errors with mean zero and variance σ^2, with no distributional assumption beyond two moments.

$$H_0 : f(x) = y^s(x) \text{ for all } x \in \Omega \quad \text{vs.} \quad H_1 : f(x) \neq y^s(x) \text{ for some } x \in \Omega$$

Validation means testing this functional hypothesis from the noisy, randomly-located sample $\{(x_i, y_i)\}_{i=1}^n$ — exactly what the circle game asked informally.

This is the model behind every slide that follows.

04 · THE AIM

Statistical validation means a p-value with a guarantee.

Two versions of the same question — over the whole domain, or over one piece of it.

1 Global validation

$$H_0 : f(x) = y^s(x) \text{ for all } x \in \Omega$$
$$H_1 : f(x) \neq y^s(x) \text{ for some } x \in \Omega$$

Existing goodness-of-fit methods can be adapted.

2 Subdomain validation

$$H_{0i} : f(x) = y^s(x) \text{ for all } x \in \Omega_i$$
$$H_{1i} : f(x) \neq y^s(x) \text{ for some } x \in \Omega_i$$

PRE-SPECIFIED PARTITION

$\Omega = \bigcup_{i=1}^{N} \Omega_i$

Existing methods largely target the global problem rather than pre-specified subdomains.

IER (Individual Error Rate) — false rejection risk for one subdomain

FWER (Family-Wise Error Rate) — risk of ≥1 false rejection across all subdomains

Identify where the model is reliable, controlling the IER locally or the FWER across all N.

Global validation



test H_0 over all of Ω at once

Subdomain validation

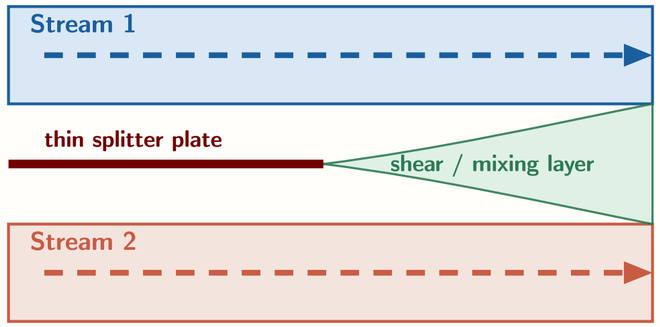


$\Omega = \bigcup_{i=1}^{N} \Omega_i$ — test H_{0i} on one piece Ω_i at a time

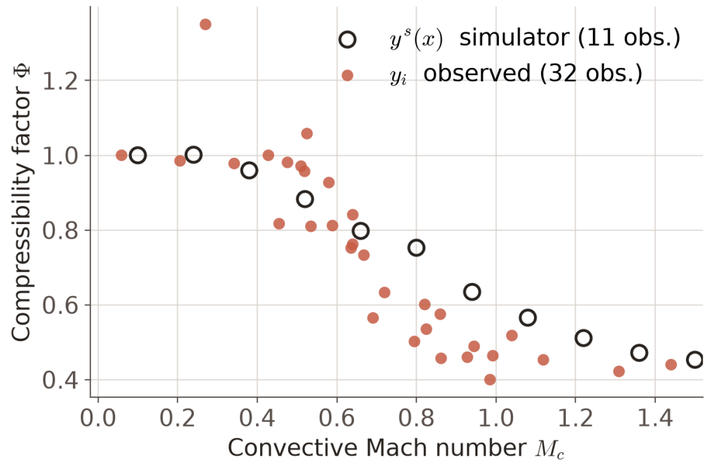
The aim is a calibrated answer, not a visual comparison.

05 · REAL EXAMPLE

A real validation example: the compressible shear layer.



The two streams remain approximately parallel, while the shear layer between them grows in thickness downstream of the splitter plate.



Black $= y^s(x)$, the simulator (11 runs). Red $= y_i$, the observed response (32 runs).

Input x

Mach number M_c

Model $y^s(x)$

simulator, 11 runs

Process $f(x)$

unknown physical relation

Data y_i

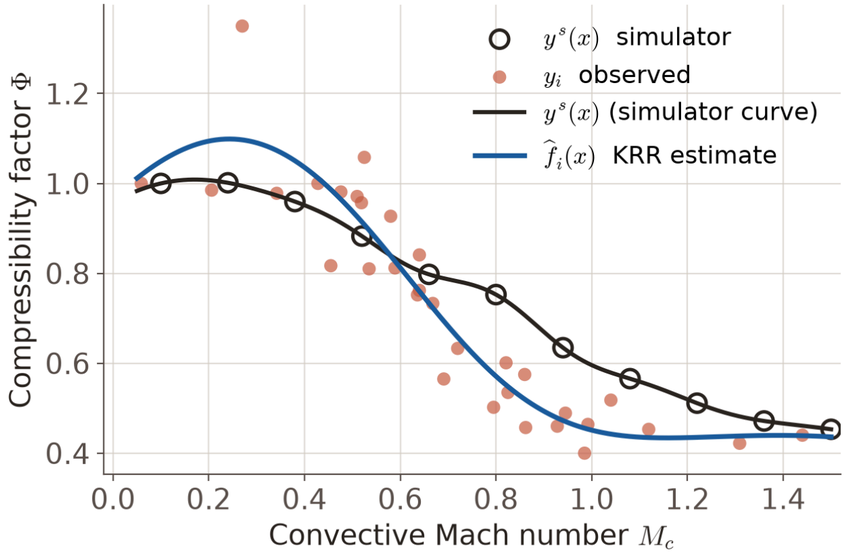
32 physical observations

Testing whether $f = y^s$ on Ω, using this one dataset.

Oberkampf & Barone (2006); Wang, Chen & Tsui (2009).

06 · STEP 1 · ESTIMATE

Smooth the noisy physical data before comparing it to the model.



Kernel ridge regression turns 32 noisy red points into a smooth estimate $\hat{f}_i(x)$.

→The black curve is $y^s(x)$ — it passes through the 11 simulator points exactly, since the model is deterministic.

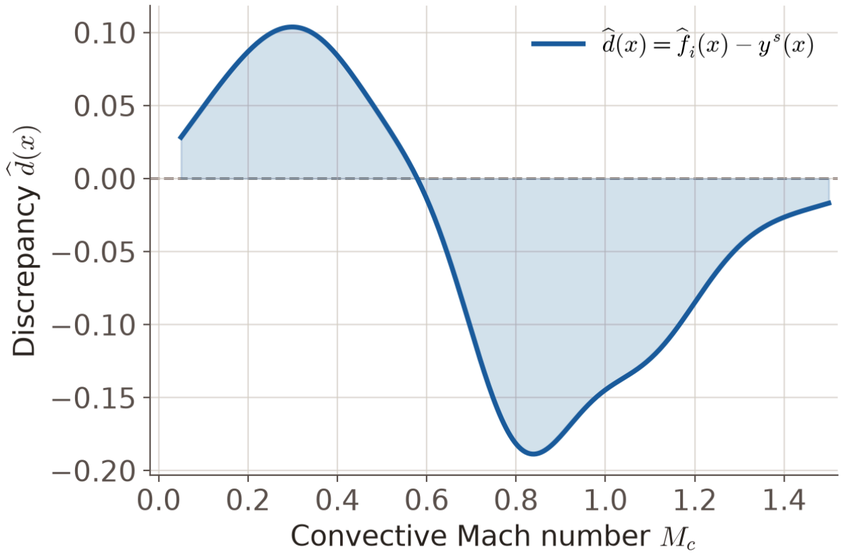
→The blue curve is $\hat{f}_i(x)$, the KRR estimate of the true physical response $f(x)$ behind the red noise, from an isotropic Matérn kernel.

→We similarly estimate the design density p and the noise scale σ by $\hat{p}$ and $\widehat{\sigma}$ — all three plug into the coefficients next.

The fitted curve is intermediate — not yet the validation result.

07 · STEP 2 · DISCREPANCY

Subtract the known model to isolate what's left unexplained.



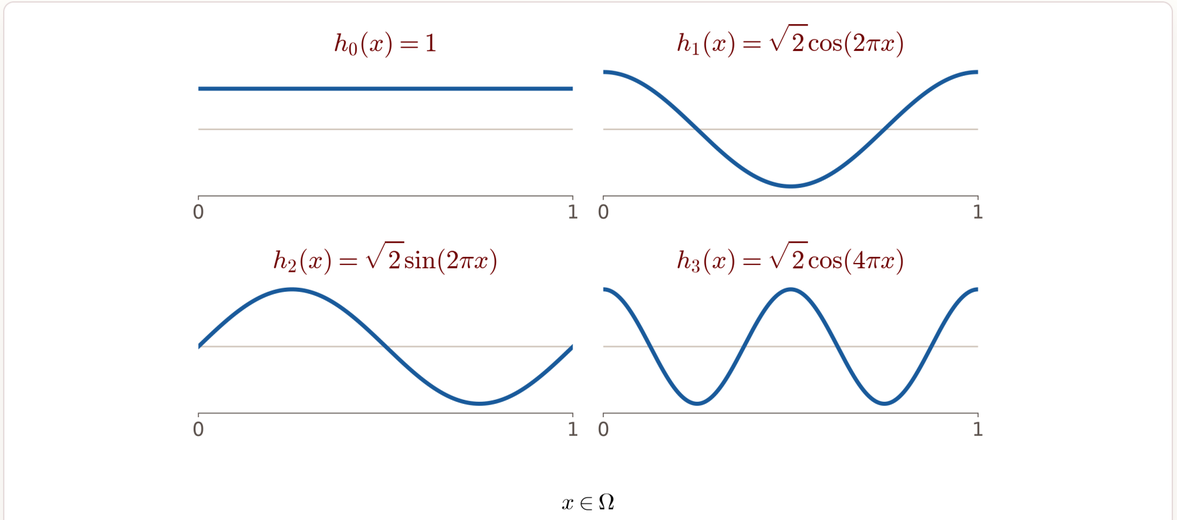
$\hat{d}(x) = \hat{f}_i(x) - y^s(x)$ is the object the null hypothesis claims is zero.

> $H_0 : d(x) = 0$ for all $x \in \Omega$
>
> If the simulator is correct, this curve should hug zero across the whole domain.

This discrepancy function is exactly what the Fourier Maximum Modulus Test (FMMT) tests.

08 · METHOD

Test the discrepancy through its coefficients, not pointwise.



$\{h_j\}_{j\geq 0}$ — an orthonormal basis of $L^2(\Omega)$.

Every square-integrable function is a sum of coefficients times basis functions.

$$g = \sum_{j=0}^{\infty} g_j\, h_j, \qquad g_j = \int_\Omega g(x)\, h_j(x)\, dx$$

g_j is a generalized Fourier coefficient of g.

Fact

$g \equiv 0$ iff $g_j = 0$ for all j — turning the functional H_0 into countably many coefficient hypotheses.

$$s(j) := \int_\Omega [f(x) - y^s(x)]\sqrt{p(x)}\, h_j(x)\, dx$$

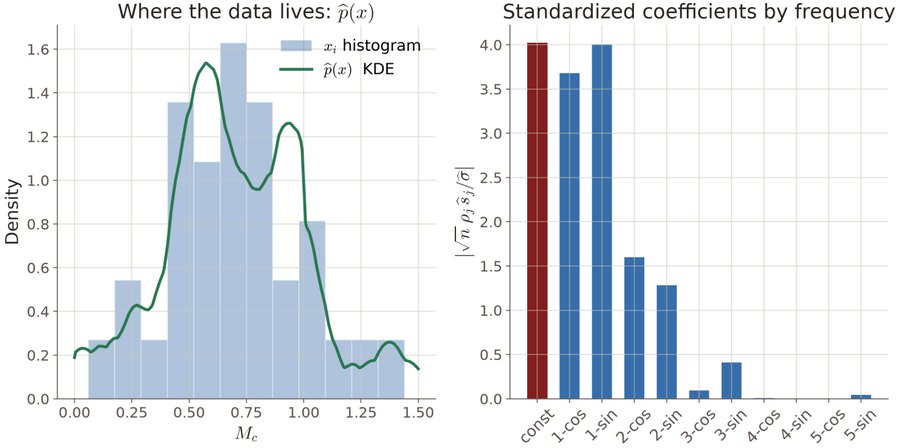
Coefficients of the discrepancy $d = f - y^s$. Goal: test $s(j) = 0$ for all j, simultaneously.

Many small, calibrated questions instead of one big vague one.

09 · STEP 3 · DECOMPOSE

Weight by density, then split into frequency directions.

The discrepancy is weighted by $\sqrt{\hat{p}(x)}$ — the input density — before it is projected onto a generalized Fourier basis.



$$\hat{s}_n(j) = \int_\Omega \hat{d}(x)\sqrt{\hat{p}(x)}\, h_j(x)\, dx$$

The estimator of $s(j)$, built from the fitted curves. The tallest (maroon) bar is the candidate maximum that step 4 turns into a p-value.

Weighting by density keeps sparse regions from being over-trusted.

10 · THEOREM

NULL LAW → P-VALUE

A joint Gaussian limit makes the maximum interpretable.

All coefficients converge together — indexed by decay weights ρ_j — not one at a time.

Theorem 1 (regularity, e.g. $\rho_j = o\big((\log(j+1))^{-1/2}\big)$): $\sqrt{n}\,\big(\rho_j(\hat{s}_n(j) - s(j))\big)_{j\geq 0} \;\Rightarrow\; N\big(0,\ \mathrm{diag}(\rho_j^2\sigma^2)_{j\geq 0}\big)$

The decay weights ρ_j are what make an infinite collection of coefficient errors converge to something usable.

Test statistic

$$T = \max_{j\geq 0} \left|\sqrt{n}\,\rho_j\,\hat{s}_n(j)/\widehat{\sigma}\right|$$

The largest standardized, decay-weighted coefficient — searching across discrepancy patterns without choosing one in advance.

Asymptotic p-value

$$\text{p-value} = 1 - \prod_{j=0}^{\infty}\left[2\Phi(T/\rho_j) - 1\right]$$

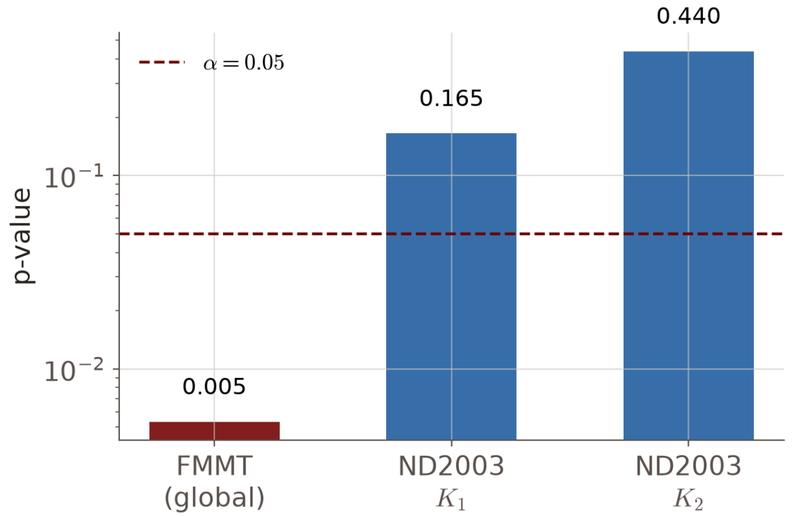
Closed-form, not a black box — the asymptotic null law of a maximum over infinitely many directions.

For a subdomain Ω_i: the same idea gives T_i from a local basis $\{h_{ij}\}$ supported on Ω_i, with the Bonferroni correction controlling the family-wise error rate across all i.

The innovation is a usable null law for a maximum over many directions.

11 · RESULT

FMMT rejects globally; a comparison baseline does not.



$$p = 0.0053$$

The theorem's null law converts $T_{\text{glob}} = 4.02$ into this p-value — below $\alpha = 0.05$.

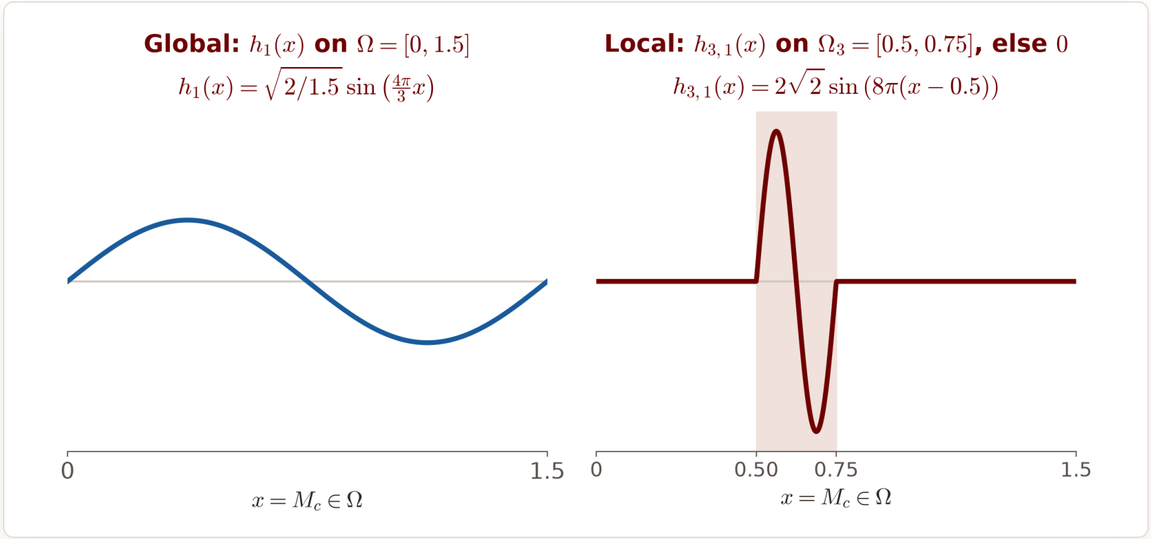
Baseline comparison

The classic two-sample test of Neumeyer & Dette (2003) — ND2003 — fails to reject (p = 0.165, 0.440). Oberkampf & Barone (2006) and Wang et al. (2012) both reject — agreeing with FMMT.

Same data, same question — FMMT detects what the baseline misses.

12 · SUBDOMAIN METHOD

Everything so far localizes — same theorem, smaller domain.



Same function, same normalization — just a shorter interval for Ω_3.

Swap the basis for one supported on Ω_i, and the whole machine carries over unchanged.

$$s_i(j) := \int_{\Omega_i} [f(x) - y^s(x)]\sqrt{p(x)}\, h_{ij}(x)\, dx$$

Same generalized Fourier coefficients — restricted to Ω_i instead of all of Ω.

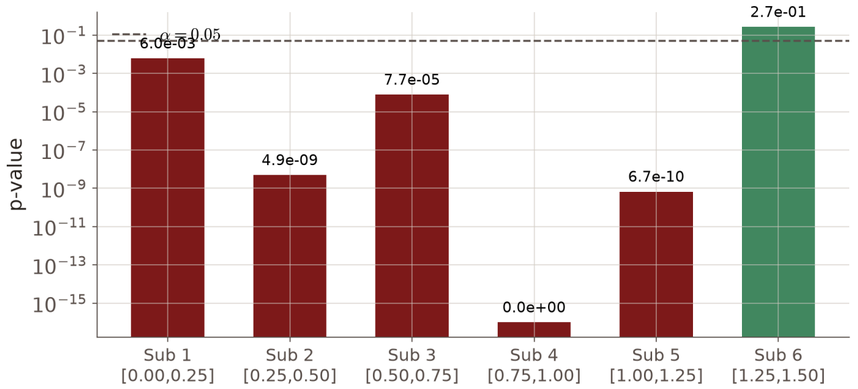
$$T_i = \max_{j \geq 0} \left|\sqrt{n}\,\rho_j\,\hat{s}_{n,i}(j)/\widehat{\sigma}\right|, \quad p_i = 1 - \prod_{j=0}^{\infty} \left[2\Phi(T_i/\rho_j) - 1\right]$$

Theorem 1 applies verbatim — its proof never used global support. Only new step: a Bonferroni correction across all N subdomains to control the FWER.

The theorem doesn't know the difference between Ω and Ω_i.

13 · SUBDOMAIN MAP

Repeating the same four steps on six subintervals turns one p-value into a map.



1	2	3	4	5	6
0.00–0.25 flagged	0.25–0.50 flagged	0.50–0.75 flagged	0.75–1.00 flagged	1.00–1.25 flagged	1.25–1.50 consistent

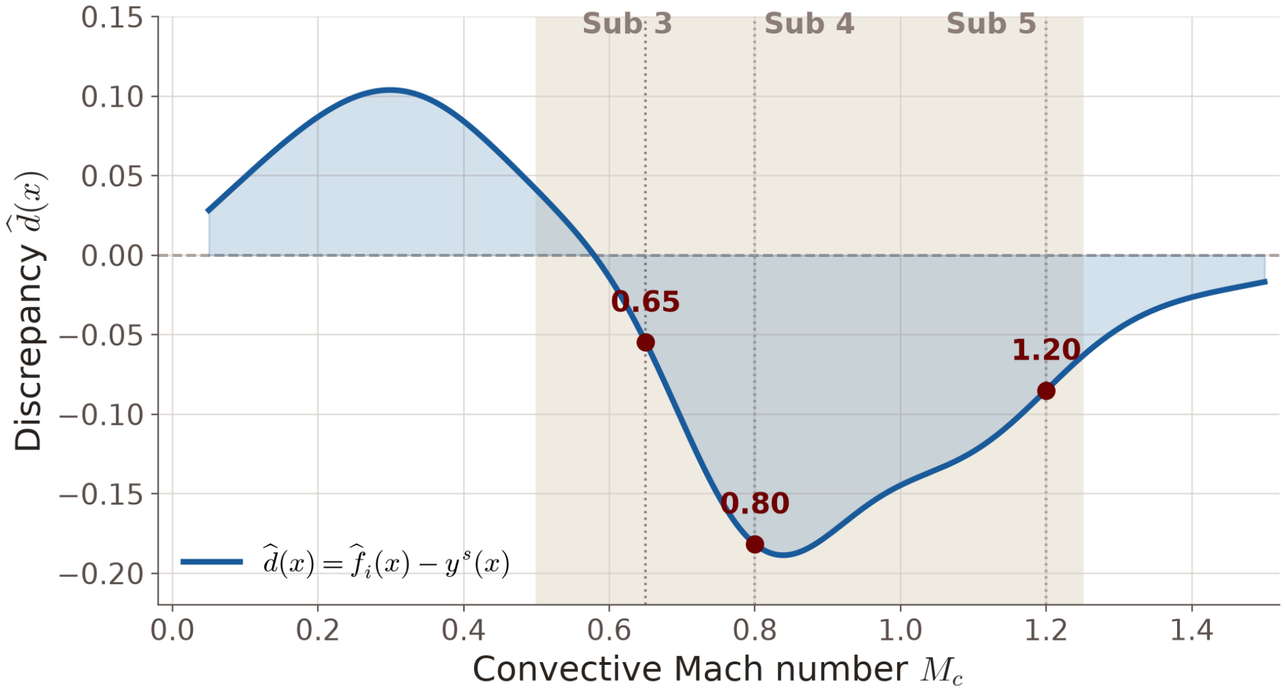
Only the highest-Mach subdomain remains statistically consistent with the simulator.

Not "the model fails everywhere" — "here is the one region it still holds."

14 · COMPARISON

Structured discrepancies explain the different decisions.

What does the discrepancy actually look like where these methods disagreed?



$M_c \approx 0.65$ — FMMT: REJECT

Prior: O&B **Not rejected** · Wang **Not rejected**

Structured discrepancy — sign changes across the interval

$M_c \approx 0.80$ — FMMT: REJECT

Prior: O&B **Reject** · Wang **Reject**

Large, unambiguous discrepancy — all three agree

$M_c \approx 1.20$ — FMMT: REJECT

Prior: O&B **Reject** · Wang *Insufficient data*

Discrepancy persists in a sparse-data region

Wang, Chen & Tsui (2009); Oberkampf & Barone (2006).

15 · CLOSE

Validation should say how strong the mismatch is, and where it does or doesn't hold.

01 We propose a **novel hypothesis testing methodology** to validate a computer model globally or in a subdomain of the input space.

02 The same machinery also applies to **two-sample function comparison** beyond model validation. Preprint: arXiv:2604.17144.

Thank you for attending!

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Q&A